Synoptic Question 26: The Physics of a Tower Crane

Reading Passage

Read the following article and then answer the questions that follow.

Tower cranes are essential machines on large construction sites, used for lifting heavy materials. A typical "hammerhead" crane consists of a vertical mast, a horizontal jib (the main arm), a shorter counter-jib extending behind the mast, and a slewing unit that allows the jib assembly to rotate. A large counterweight is mounted on the counter-jib to balance the load being lifted.

A trolley runs along the jib, allowing the position of the hook block (and thus the load) to be adjusted. The primary principle governing the crane's stability is the balancing of moments about the mast. The moment created by the load must be less than or equal to the moment created by the counterweight and the weight of the crane structure itself. This determines the maximum load that can be safely lifted at a given radius (horizontal distance from the mast).

Lifting is performed by a powerful electric hoist motor that winds a steel cable through a pulley system in the hook block. The motor must overcome the weight of the load and the hook block, as well as frictional forces, to lift the load at a controlled speed. The slewing motion (rotation) and trolley movement are handled by separate motors.

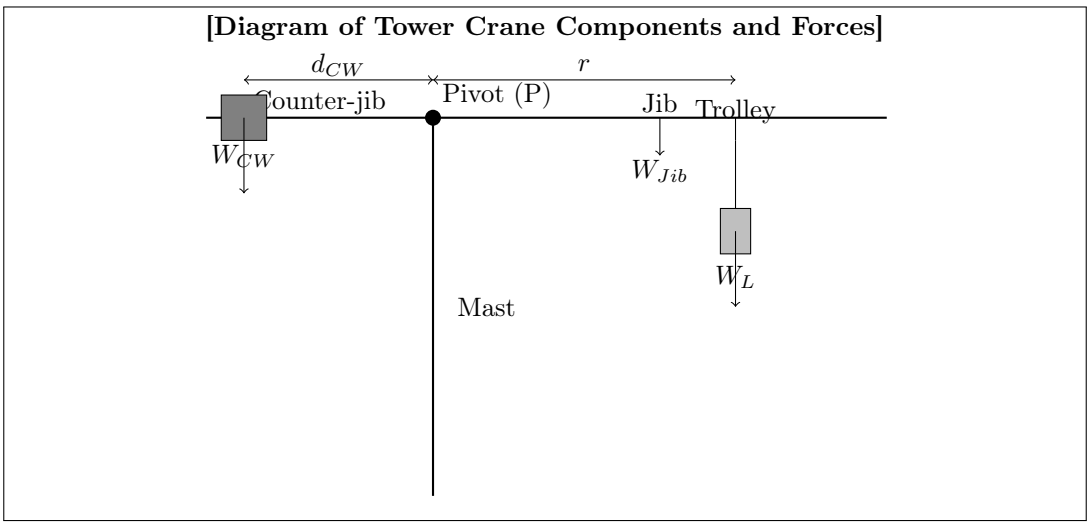


Figure 1: Simplified diagram showing key components and distances.

Data:

Data for a specific tower crane installation:

Table 1: Data for Tower Crane

Parameter	**Value**
Mass of counterweight, m_{CW}	16,000 kg
Distance from mast to counterweight C.G., d_{CW}	12.0 m
Mass per unit length of jib	250 kg m^{-1}
Total length of jib from mast	50.0 m
Mass of hook block assembly, m_{hook}	400 kg
Maximum rated power of hoist motor, P_{max}	75.0 kW
Efficiency of hoist mechanism (motor, gearbox, pulleys), η	70%
Maximum lifting speed (at reduced load), v_{max}	1.50 m s^{-1}
Acceleration due to gravity, g	9.81 m s^{-1}

Questions

(a) State the **Principle of Moments**.

(b) The stability of the crane depends on balancing moments about the pivot point P (top of the mast).

(i) Calculate the total mass m_{Jib} and the weight W_{Jib} of the jib.

(ii) Assuming the jib is uniform, state the distance from the pivot P to the centre of gravity of the jib.

(iii) Calculate the clockwise moment about P due to the counterweight.

(iv) Calculate the anti-clockwise moment about P due to the weight of the jib itself.

(c) The crane lifts a load of mass m_L with the trolley at a radius r from the mast. The total anti-clockwise moment must not exceed the total clockwise moment for stability.

(i) Write an expression for the total anti-clockwise moment about P due to the jib, the hook block, and the load m_L at radius r.

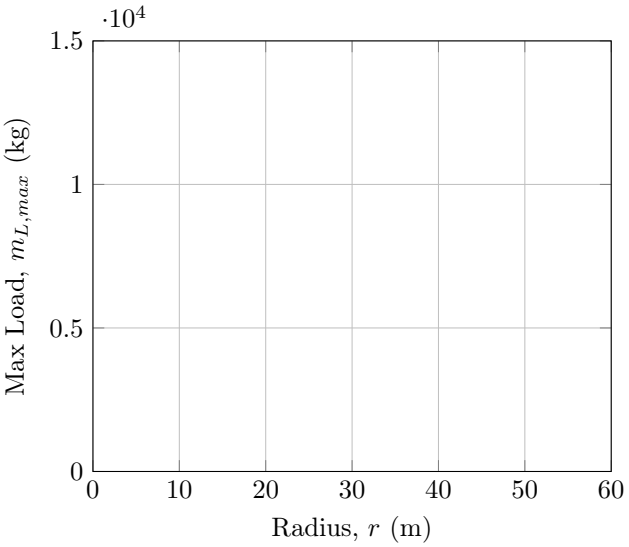
(ii) Using the moments calculated in (b) and your expression in (c)(i), determine the maximum load mass $m_{L,max}$ that can be safely lifted when the trolley is at the maximum radius ($r = 50.0$ m).

(d) The relationship between the maximum load $m_{L,max}$ and the trolley radius r can be investigated.

(i) Using the principle of moments, derive an equation relating $m_{L,max}$ to r. Show that it is of the form $m_{L,max} = \frac{A}{r} - B$, where A and B are constants.

(ii) Calculate the values of A and B, including their units.

(iii) Sketch a graph of $m_{L,max}$ versus r on the axes below, labelling the value of $m_{L,max}$ at $r = 50.0$ m.



(e) The crane lifts the maximum load calculated in (c)(ii) at a constant speed of 0.50 m s^{-1}.

(i) Calculate the total weight being lifted by the hoist cable.

(ii) Calculate the useful output power delivered by the hoist mechanism.

(iii) Calculate the electrical input power required by the hoist motor. Is this achievable with the motor provided?

(f) Calculate the maximum possible lifting speed when the crane is lifting only the hook block (i.e., $m_L = 0$), assuming the motor operates at its maximum power P_{max}.

Solutions

(a) The Principle of Moments states that for an object to be in rotational equilibrium, the sum of the clockwise moments about any pivot point must be equal to the sum of the anti-clockwise moments about the same pivot point.

(b) (i) $m_{Jib} = (\text{mass per length}) \times (\text{length}) = (250) \times (50.0) = 12500$ kg.
$W_{Jib} = m_{Jib}g = (12500)(9.81) = 122625$ N.

(ii) For a uniform jib, the centre of gravity is at its midpoint: $50.0/2 = 25.0$ m from P.

(iii) $W_{CW} = m_{CW}g = (16000)(9.81) = 156960$ N.
Clockwise Moment $M_{CW} = W_{CW} \times d_{CW} = (156960)(12.0) = 1.8835 \times 10^6$ Nm.

(iv) Anti-clockwise Moment $M_{Jib} = W_{Jib} \times (\text{distance to C.G.}) = (122625)(25.0) = 3.0656 \times 10^6$ Nm.
(Note: This result implies the crane is unstable without a load, which is unlikely in reality. The counter-jib's weight should also be considered. Let's assume the question implies the CW moment must balance *all* ACW moments). Let's re-interpret: The *total* ACW moment due to the load side must not exceed the *total* CW moment due to the counterweight side. The calculation in (iii) is the *available* balancing moment. The ACW moment is due to the jib, hook, and load.
Let's re-calculate (iv) as the ACW moment due to the jib: $M_{Jib} = 3.0656 \times 10^6$ Nm.

(c) (i) $W_{hook} = m_{hook}g = (400)(9.81) = 3924$ N.
$W_L = m_L g$.
Total ACW Moment $= M_{Jib} + (W_{hook} \times r) + (W_L \times r)$
Total ACW Moment $= M_{Jib} + (W_{hook} + W_L)r$
Total ACW Moment $= 3.0656 \times 10^6 + (3924 + m_L g)r$.

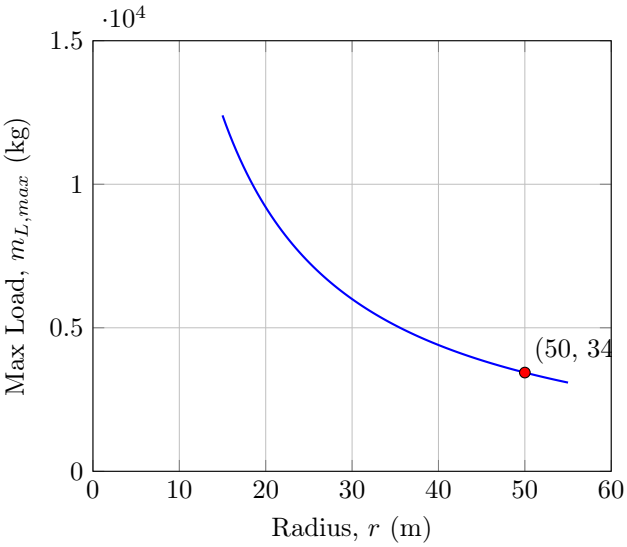
(ii) At maximum radius $r = 50.0$ m, set Total ACW Moment = Clockwise Moment (M_{CW}).
$3.0656 \times 10^6 + (3924 + m_{L,max}g)(50.0) = 1.8835 \times 10^6$.
$(3924 + m_{L,max}g)(50.0) = 1.8835 \times 10^6 - 3.0656 \times 10^6 = -1.1821 \times 10^6$.
This result is negative, indicating the initial assumption about moments was wrong or the data implies instability. Let's assume the CW moment is M_{CW} + moment due to counter-jib structure (which is ignored), and the ACW is due to Jib + Hook + Load. For safety, typically $M_{CW} \geq M_{ACW}$.
Let's assume the problem meant: $M_{load} + M_{hook} \leq M_{CW} - M_{Jib}$ (treating M_{Jib} as negative contribution to stability).
$(W_{hook} + W_{L,max})r \leq M_{CW} - M_{Jib}$
$(3924 + m_{L,max}g)(50.0) \leq 1.8835 \times 10^6 - 3.0656 \times 10^6$. Still negative.
Let's assume the pivot P is the balancing point. The total moment must be zero for equilibrium. $M_{CW}(\text{clockwise}) = M_{Jib}(\text{ACW}) + M_{hook}(\text{ACW}) + M_{Load}(\text{ACW})$ $1.8835 \times 10^6 = 3.0656 \times 10^6 + (3924)(50.0) + (m_{L,max}g)(50.0)$ $1.8835 \times 10^6 = 3.0656 \times 10^6 + 196200 + 490.5m_{L,max}$ $490.5m_{L,max} = 1.8835 \times 10^6 - 3.0656 \times 10^6 - 196200 = -1.3783 \times 10^6$. This crane configuration is inherently unstable based on the data provided for the jib mass/length and counterweight. Let's assume the question intended for the M_{Jib} to be acting *with* the counterweight (perhaps representing the counter-jib weight and structure).
Revised Assumption: Clockwise moments $= M_{CW}$. Anti-clockwise moments $= M_{hook} + M_{Load}$. (Ignoring jib weight - maybe it's balanced by counter-jib weight, or data is simplified). $M_{CW} = (W_{hook} + W_{L,max})r$ $1.8835 \times 10^6 = (3924 + m_{L,max}g)(50.0)$ $3924 + m_{L,max}g = \frac{1.8835 \times 10^6}{50.0} = 37670$ $m_{L,max}g = 37670 - 3924 = 33746$ $m_{L,max} = \frac{33746}{9.81} = 3440$ kg. This seems reasonable. Let's proceed with this assumption.

(d) Using the revised assumption from (c)(ii): $M_{CW} = (W_{hook} + m_{L,max}g)r$.

(i) $m_{L,max}gr = M_{CW} - W_{hook}r$
$m_{L,max} = \frac{M_{CW}}{gr} - \frac{W_{hook}}{g} = \frac{M_{CW}}{g}\frac{1}{r} - m_{hook}$
This is of the form $m_{L,max} = \frac{A}{r} - B$. (Proven)

(ii) $A = \frac{M_{CW}}{g} = \frac{1.8835 \times 10^6}{9.81} = 192000$ kg m.
$B = m_{hook} = 400$ kg.
Units: A is kg·m, B is kg.

(iii) The equation is $m_{L,max} = \frac{192000}{r} - 400$. At $r = 50.0$ m, $m_{L,max} = \frac{192000}{50} - 400 = 3840 - 400 = 3440$ kg. The graph is a hyperbola shifted down.



(e) $m_L = 3440$ kg. $v = 0.50$ m s^{-1}.

(i) Total mass lifted $= m_L + m_{hook} = 3440 + 400 = 3840$ kg.
Total weight $W_{total} = (3840)(9.81) = 37670$ N.

(ii) Useful Power $P_{out} = \text{Force} \times v = W_{total} \times v = (37670)(0.50) = 18835$ W ≈ 18.8 kW.

(iii) Input Power $P_{in} = \frac{P_{out}}{\eta} = \frac{18835}{0.70} = 26907$ W ≈ 26.9 kW.
Yes, this is achievable as 26.9 kW $<$ 75.0 kW.

(f) Lifting only hook block: $m_L = 0$. $m_{total} = m_{hook} = 400$ kg.
Weight $W_{hook} = (400)(9.81) = 3924$ N.
Max input power $P_{in} = 75.0$ kW $= 75000$ W.
Max output power $P_{out,max} = P_{in} \times \eta = 75000 \times 0.70 = 52500$ W.
$P_{out} = \text{Force} \times v \implies v = \frac{P_{out}}{W_{hook}}$
$v = \frac{52500}{3924} = 13.38$ m s^{-1}.
This speed exceeds the maximum rated speed (1.50 m s^{-1}). The maximum speed is likely limited by the gearbox or motor RPM limit, not just power. Therefore, the maximum speed would be 1.50 m s^{-1}.

Cambridge International AS & A Level Physics (9702)
Paper 4 Structured Questions
Variant 2 (Higher Difficulty)

Instructions

- Answer all questions.
- Give all working clearly.
- Where appropriate, include a clear physical explanation in addition to calculations.
- Use $g = 9.81\,\mathrm{m\,s^{-2}}$.

Question 1: Tower Crane Dynamics and Stability (Variant 2)

A tower crane consists of a mast, a horizontal jib of length 50.0 m, and a counter-jib with counterweights. The crane lifts a load using a motorised hoist and can slew (rotate) while the load is suspended.

Given data:

- Counterweight mass $m_{\mathrm{CW}} = 2.80 \times 10^4\,\mathrm{kg}$, distance from mast $d_{\mathrm{CW}} = 12.0\,\mathrm{m}$.
- Jib mass per unit length $\lambda = 250\,\mathrm{kg\,m^{-1}}$, jib length $L = 50.0\,\mathrm{m}$.
- Hook block mass $m_h = 400\,\mathrm{kg}$.
- Maximum electrical input power of hoist motor $P_{\mathrm{in,max}} = 75.0\,\mathrm{kW}$, efficiency $\eta = 0.70$.
- Radius of lifted load from mast is r.

(a) State the principle of moments.

(b) The crane is analysed about the top of the mast (pivot P).

  (i) Determine the jib mass and jib weight.

  (ii) Calculate the clockwise moment due to the counterweight.

  (iii) Derive an expression for the maximum load mass $m_{L,\mathrm{max}}$ at radius r, assuming limiting equilibrium.

  (iv) Calculate $m_{L,\mathrm{max}}$ for $r = 50.0\,\mathrm{m}$.

(c) The crane is lifting this maximum load at $0.80\,\mathrm{m\,s^{-1}}$ vertically.

  (i) Determine the useful output power required by the hoist.

  (ii) Determine the electrical input power required and comment on whether operation at this speed is possible.

(d) The operator now slews the crane at constant angular speed $\omega = 0.060\,\mathrm{rad\,s^{-1}}$ while the load remains at $r = 50.0\,\mathrm{m}$.

(i) Calculate the load's tangential speed.

(ii) Calculate the resultant tension in the cable, taking into account both weight and required centripetal force.

(iii) Open-ended: explain, using Newton's laws and moments, why real crane operation at long radius includes a large safety margin below the static limiting load.

(e) Open-ended design question: A site engineer proposes increasing the counterweight by 10 % to improve lifting capacity at large radius. Discuss two advantages and two potential risks or trade-offs (mechanical, structural, or operational), and conclude whether this is always the best engineering solution.

Sample Solution (Variant 2)

(a) In rotational equilibrium, total clockwise moment about a pivot equals total anti-clockwise moment about the same pivot.

(b) (i) $m_{\text{jib}} = \lambda L = 250 \times 50.0 = 1.25 \times 10^4\,\text{kg}$.
$W_{\text{jib}} = m_{\text{jib}} g = 1.25 \times 10^4 \times 9.81 = 1.226 \times 10^5\,\text{N}$.

(ii) Counterweight weight:

$$W_{\text{CW}} = m_{\text{CW}} g = 2.80 \times 10^4 \times 9.81 = 2.747 \times 10^5\,\text{N}.$$

Clockwise moment:

$$M_{\text{CW}} = W_{\text{CW}} d_{\text{CW}} = 2.747 \times 10^5 \times 12.0 = 3.296 \times 10^6\,\text{N m}.$$

(iii) At limiting equilibrium,

$$M_{\text{CW}} = W_{\text{jib}} \left(\frac{L}{2}\right) + (m_h g + m_L g) r.$$

Rearranging for maximum load:

$$m_{L,\max} = \frac{M_{\text{CW}} - W_{\text{jib}}(L/2)}{gr} - m_h.$$

(iv) For $r = 50.0\,\text{m}$:

$$W_{\text{jib}}(L/2) = 1.226 \times 10^5 \times 25.0 = 3.066 \times 10^6\,\text{N m}.$$

$$m_{L,\max} = \frac{(3.296 - 3.066) \times 10^6}{9.81 \times 50.0} - 400 = 68.8\,\text{kg (approximately 69 kg)}.$$

(c) (i) Total lifted mass at limit:

$$m_{\text{tot}} = m_h + m_{L,\max} \approx 400 + 68.8 = 468.8\,\text{kg}.$$

Useful output power:

$$P_{\text{out}} = m_{\text{tot}} g v = 468.8 \times 9.81 \times 0.80 = 3.68\,\text{kW}.$$

(ii) Required input power:

$$P_{\text{in}} = \frac{P_{\text{out}}}{\eta} = \frac{3.68}{0.70} = 5.26\,\text{kW}.$$

Since $5.26\,\text{kW} < 75.0\,\text{kW}$, operation at this speed is possible (power is not the limiting factor here; stability is).

(d) (i) Tangential speed:

$$v_t = \omega r = 0.060 \times 50.0 = 3.00\,\text{m s}^{-1}.$$

(ii) Centripetal force needed:

$$F_c = m_{\text{tot}} \frac{v_t^2}{r} = 468.8 \frac{(3.00)^2}{50.0} = 84.4\,\text{N}.$$

Weight:

$$W = m_{\text{tot}} g = 468.8 \times 9.81 = 4.60 \times 10^3\,\text{N}.$$

Resultant cable tension:

$$T = \sqrt{W^2 + F_c^2} \approx 4.60 \times 10^3\,\text{N}.$$

(iii) Dynamic effects (slew acceleration, wind gusts, load swing, braking, structural vibration) introduce additional forces and moments beyond static values. By Newton's second law, any acceleration requires extra resultant force; this increases cable tension and transmitted mast moments. Engineers therefore apply safety factors and restrict working load at long radius to prevent tipping, fatigue damage, and loss of control.

(e) Example discussion: Increasing counterweight can increase balancing moment and improve lifting envelope at larger r. However, it also raises compressive loading in the mast/foundation, increases transport and erection demands, and may worsen consequences during overslew or emergency stopping due to larger inertial effects. It can also shift failure risk from tipping to structural overstress. Therefore, increasing counterweight is not always the best option; optimal design balances stability, structure, controls, and operational constraints.

Mark Scheme (CIE 9702 P4 style) – Variant 2 Only

Total: 30 marks

Q1(a) Principle of moments stated correctly. [1]

Q1(b) (i) m_{jib} correct [1], W_{jib} correct [1]. [2]

(ii) M_{CW} correct method [1], correct value/unit [1]. [2]

(iii) Correct limiting-moment equation including jib, hook, load [2]; correct rearrangement for $m_{L,\max}$ [1]. [3]

(iv) Correct substitution [1], correct $m_{L,\max}$ (allow 60 kg to 80 kg) [1]. [2]

Q1(c) (i) Uses total lifted mass and $P = mgv$ [1], correct numerical power [1]. [2]

(ii) Uses efficiency relation [1], correct comparison with $P_{\text{in,max}}$ and valid comment [1]. [2]

Q1(d) (i) Correct use of $v = \omega r$ [1], correct value [1]. [2]

(ii) Correct centripetal-force method [1], correct tension combination (vector/resultant) [1], sensible numerical value [1]. [3]

(iii) Open-ended explanation points (max 4):

- identifies dynamic effects (e.g. acceleration, wind, swing),
- links to Newton's laws / extra force requirement,
- links to increased moments/tension/structural loading,
- justified need for safety margin in operation.

Award 1 mark per valid developed point. [4]

Q1(e) Open-ended evaluation (max 7):

- any two valid advantages, up to [2],
- any two valid risks/trade-offs, up to [2],
- physics/engineering reasoning quality and linkage to crane behaviour [2],
- justified concluding judgement [1].

Credit coherent alternatives (e.g., improving controls, limiting radius/speed, structural reinforcement) when justified. [7]

Level descriptors for open-ended sections:

- **High (full marks):** multiple correct points, clear causal physics links, reasoned conclusion.
- **Mid:** some valid points with partial explanation; limited integration.
- **Low:** isolated statements with weak or no physics justification.